

AI Traffic Impact and Token/Intent-Based Communication

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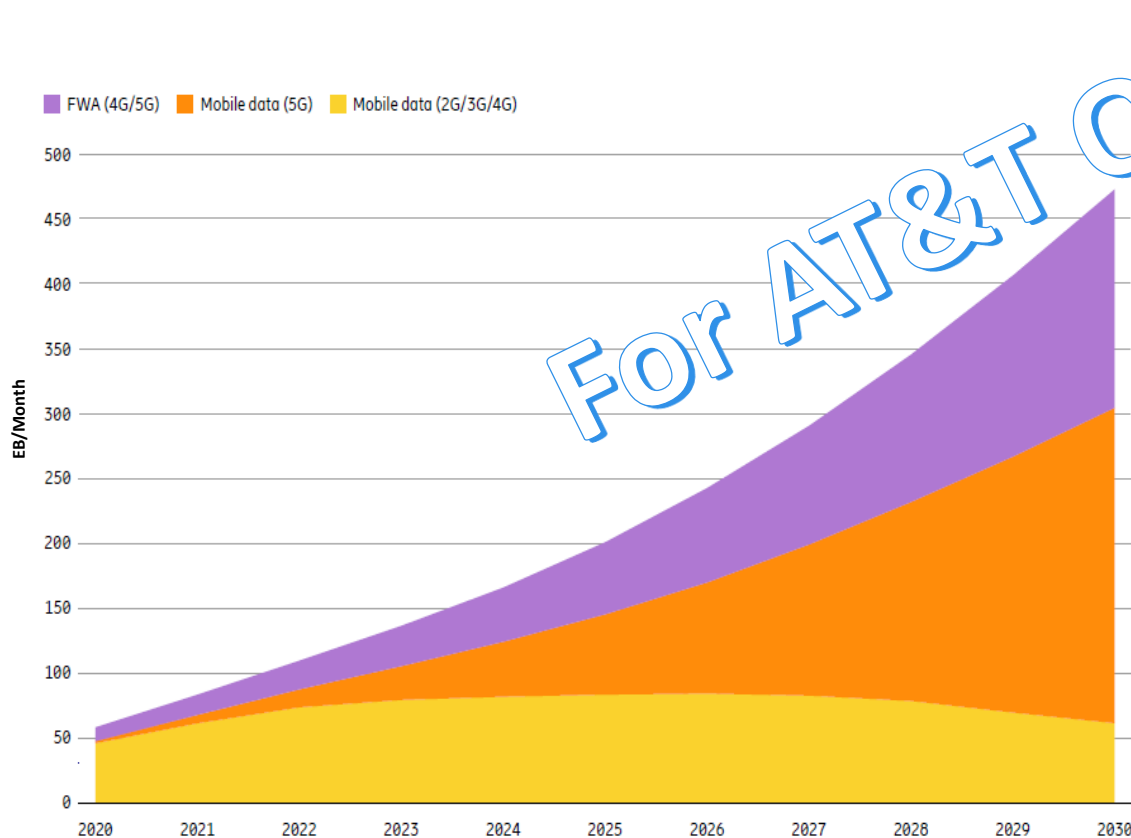
Technology Enablers – Research Studies and 3GPP Perspective

Oct 2, 2026

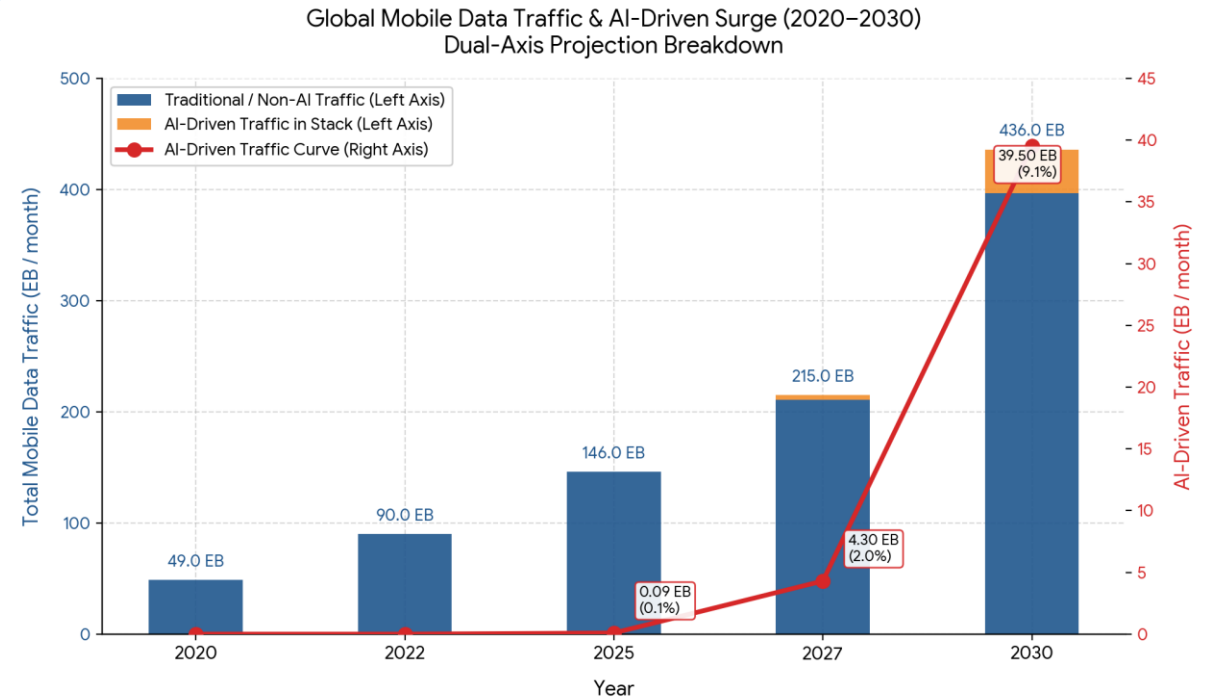
AI and Token-Based Communication

Traffic Growth Driven by Emerging AI Applications

- Overall mobile data traffic is projected to grow $\approx 10\times$ from 2020 $\rightarrow$ 2030



Source: Ericsson Mobile Report 2024



Source: Compiled from ABI research, GSMA reports

AI-Intensive Applications & Verticals

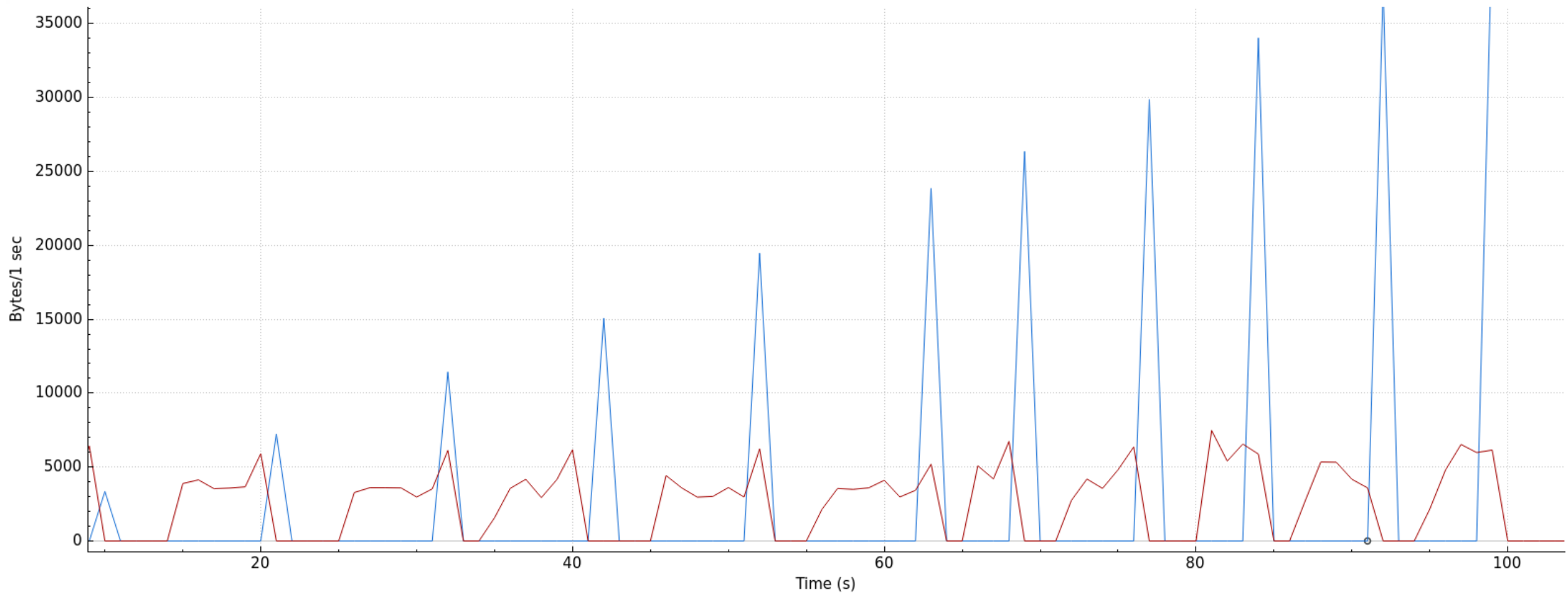
Use-Case	UL/DL Profile (typical)	Why it explodes
AR/AI glasses (e.g., mixed-reality streaming, eye-tracking)	UL $\approx$ 5 Mbps sustained; DL $\approx$ 10 Mbps bursty	Constant high-rate sensor upload (camera, IMU) & low-latency feedback
Smartphone AI assistance (voice, text, multimodal)	UL $\approx$ 500 KB per turn; 5-10 turns per session; UL context grows as the conversation deepens	Each turn carries not just raw audio but embeddings, image patches, scene graphs
On-device LLM prompting with cloud inference	UL $\approx$ 2-10 KB per token (embedding) $\times$ N-turns	Multi-turn conversation $\rightarrow$ cumulative UL payload can exceed tens of MB per session
Real-time translation / live captioning	DL $\approx$ tiny “token” packets (10-200 B) at 10-100 Hz	Per-token latency dominates perceived quality $\rightarrow$ high packet-rate, low-size traffic

FOU

Issue-1: UL Context Growth in Multi-Turn Cloud-LLM Sessions

- Due to the nature of state-less cloud model, without shared state, information must be re-transmitted → **UL grows linearly with conversation depth**

Wireshark I/O Graphs: capture_20260611_174204_gemini_chat_multiturn.pcap



Issue-1: Observations & Analysis

Root cause:

- **History resend:** cloud LLM calls are often stateless
- **Progressive context enrichment:** later turns attach more metadata
- **Application-level recovery:** extra UL bursts due to ambiguity/failures triggering resends

Network/UE impact:

- **UL becomes capacity + latency bottleneck** for AI assistant usage on smartphones
- **Increased grant/control overhead** and jitter from frequent UL bursts
- **Higher UE energy/thermal load** due to repeated uplink activity

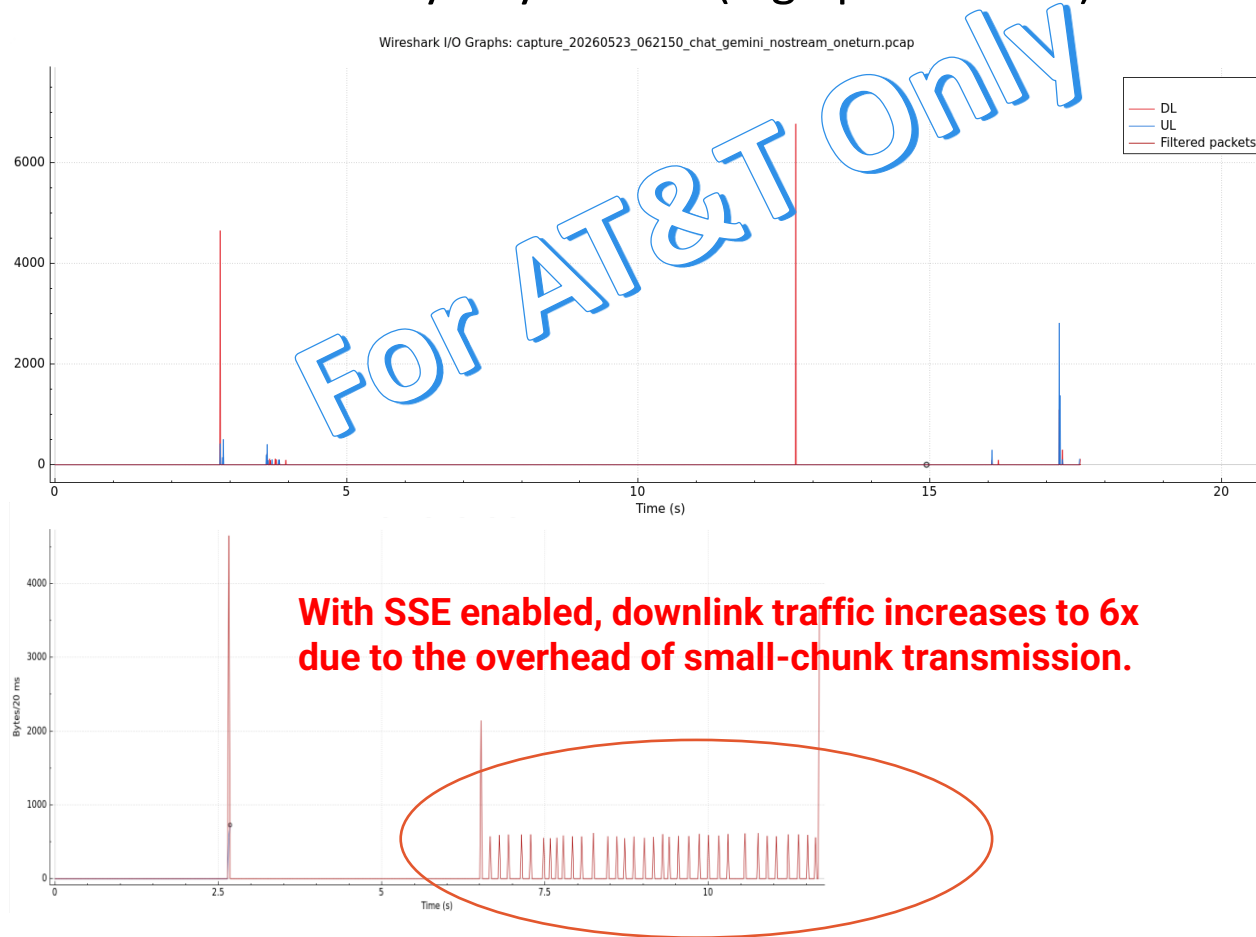


6G AI-Native implications :

- **Stateful semantic session** (send deltas vs full history)
- **UL burst-optimized** access
- **Semantic compression** (quantized embeddings, token ID + residual)
- **Cross-layer hints** (token priority / utility / deadline) for token-aware scheduling

Issue-2: Downlink Overhead Inflation for Small Packets

- Switching Gemini chat from buffered response to SSE (Server-Sent Events) streaming turns DL from few bursts into many tiny chunks (high packet rate)



AI Traffic Metrics

Latency (mean): 9.626s
Latency (p95): 9.626s
Success rate: 100.0%
Request bytes: 182
Response bytes: 12240
UL/DL ratio: 0.015
Burst P/M: 1.00

Latency (mean): 9.136s
Latency (p95): 9.136s
Success rate: 100.0%
Request bytes: 198
Response bytes: 74264
UL/DL ratio: 0.003
TTFT (p95): 9.115s
TTLT (p95): 9.136s
Stall rate: 0.000
Burst P/M: 1.00

DL grows ~6x

Issue-2: Observations & Analysis

- Streaming improves perceived responsiveness, but **inflates downlink traffic dramatically due to small-chunk overhead**

Root cause:

- App/HTTP/SSE framing (event format + chunking)
- Transport/security (TCP/QUIC + TLS record framing + ACK behavior)
- RAN stack (PDCP/RLC/MAC headers, scheduling control, padding)

Network/UE impact:

- Spectral efficiency drops (headers dominate payload for 10–200B chunks)
- Scheduler load increases (more grants/PDUs per response)
- Battery hit on UE (ex. DRX inefficiency)

6G AI-Native implications:

- Token bundling/aggregation (send N tokens per transmission opportunity)
- Ultra-light header compression optimized for token streams
- Token-aware QoS (deadline/utility) for low-jitter streaming

Trends to Alleviate AI Traffic Impact

1 On-Device AI

- ✓ Minimal Over-The-Air transmission
- ✓ Local audio/vision inference

Local processing on smart device
NPUs executing Small Language
Models (SLMs) on-chip

Reducing RAN traffic

2 Tokenization & Semantic Coding

- ✓ 10x – 100x Data Compression
- ✓ Low-bitrate task communications

Instead of transmitting raw video and
audio samples over RAN, extracting
intent, tokens, and feature maps

No redundant capacity
consumption

3 Edge AI / Split Inference

- ✓ Redesign token transport in 6G
- ✓ Sub-10ms response loops

Split model across
device-edge-cloud to keep only
high-utility tokens on the radio

Optimizing system utilization for
mission critical AI applications

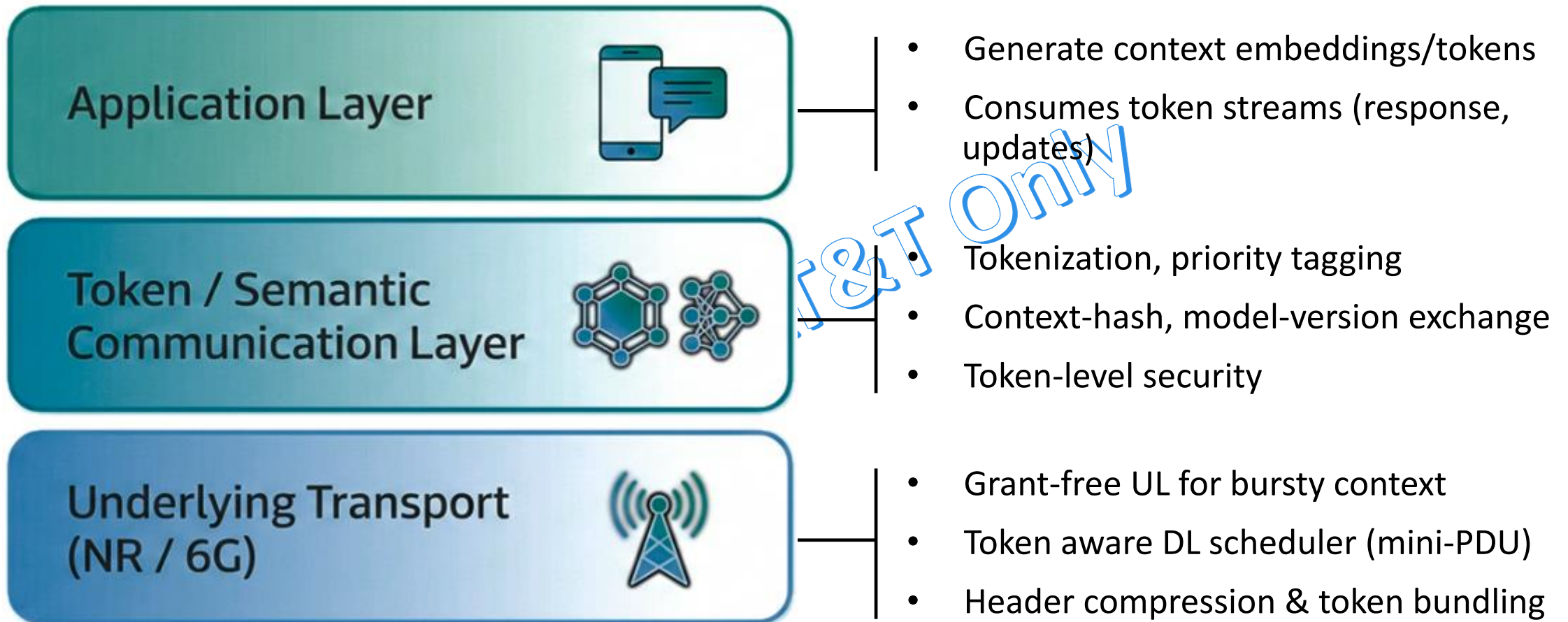
Challenges & Open Research Directions (1/2)

Core challenge	Root cause	R&D directions for 6G AI-native design
1) UL context burst growth (multi-turn)	Stateless cloud inference and insufficient shared session state → resend history/context; multi-modal enrichment over time	• Stateful semantic sessions (context hash, shared memory checkpoints, delta-context upload); • Grant-free UL for frequent bursts; • Context-aware UL scheduling (deadline/utility hints from AI runtime); • Embedding quantization + “token ID + residual”
2) DL small-payload streaming overhead (SSE)	Many micro-chunks → repeated overhead across HTTP/QUIC/TLS + PDCCP/RLC/MAC + scheduler granularity	• Token bundling/aggregation at PDCCP/RLC/MAC; • Token-optimized header compression; • Mini-slot low-jitter DL scheduling;
3) Semantic QoS (tokens are unequal importance)	Current QoS models are bit-centric (GBR/5QI) and cannot express “utility”	• Utility-weighted QoS model (token class, deadline, confidence, etc.); • Token-aware scheduler in UE + gNB; • Selective reliability (intent/action vs. info. tokens)

Challenges & Open Research Directions (2/2)

Core challenge	Root cause	R&D directions for 6G AI-native design
4) Semantic noise / model & context mismatch	Token meaning depends on model version, tokenizer, context window, tool state	• Model/version negotiation + context alignment (context hash, tokenizer ID); • Semantic error detection (inconsistency checks) and semantic recovery (query/regenerate vs retransmit)
5) Reliability vs latency tradeoff (token streams)	Classical HARQ/ARQ retransmits bits; for tokens, regeneration or inference may be cheaper/faster	• Semantic-HARQ policies; • Deadline-aware retransmission and partial delivery (coarse-to-fine tokens)
6) UE energy / thermal constraints	Frequent UL bursts + frequent DL wakeups reduce DRX benefits; on-device AI + modem concurrency heats device	• Energy-aware token admission control (utility-per-Joule); • DRX/token co-design (bundle tokens to align with wake cycles); • Dynamic split (on-device vs edge vs cloud) based on link/thermal state

Token-Based Communication – Conceptual Stack

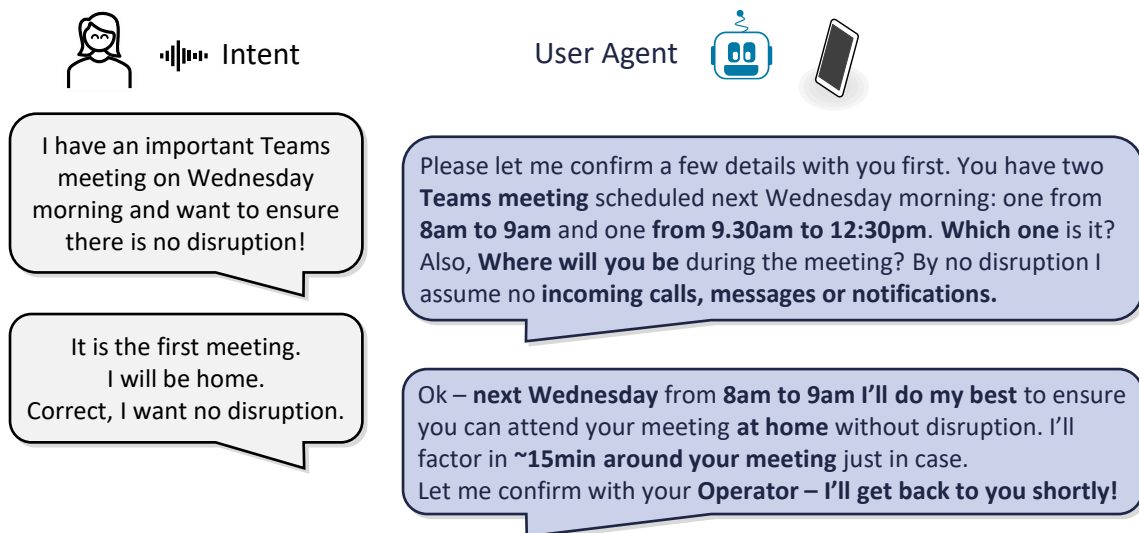


Architecture Options & Required 3GPP Support

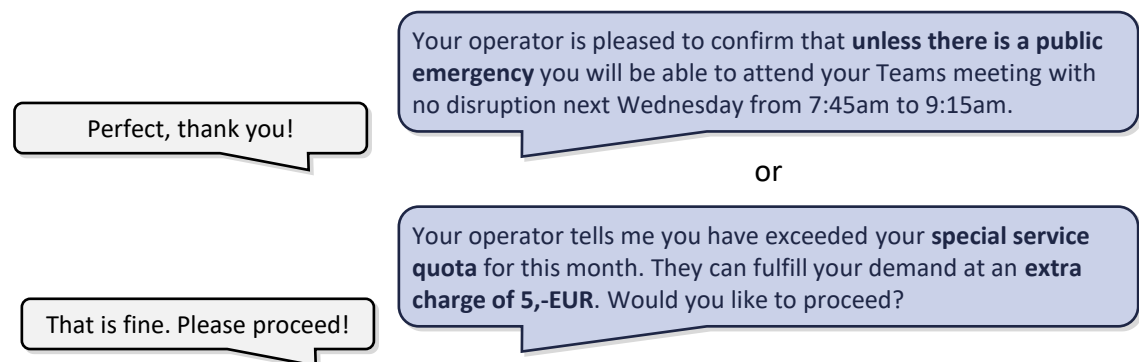
Option	Description	3GPP Feature Needed
A. Traditional Transport + Token-Layer	Tokenization lives in the app, but packets travel over existing NR-MAC/PDCP unchanged	None (baseline) – can be prototyped now
B. Token-Aware Transport	Dedicated Token Transport Channel with semantic QoS (utility-weighted scheduling, mini-slot UL, token-level HARQ)	New 6G Specs Protocol Stack Features, e.g., • RRC: SemanticSessionInfo, ... • SemanticQoS Profile (utility, latency, reliability) • PDCP/RLC: token bundling, header compression, ... • MAC: mini-slot, grant-free UL, ... • PHY: UL boosting, unequal FEC, ...

Intent-Based Communication

Monetizing Better UXP – User Intent Use Case



1. Expressing and understanding intent: collect and compile



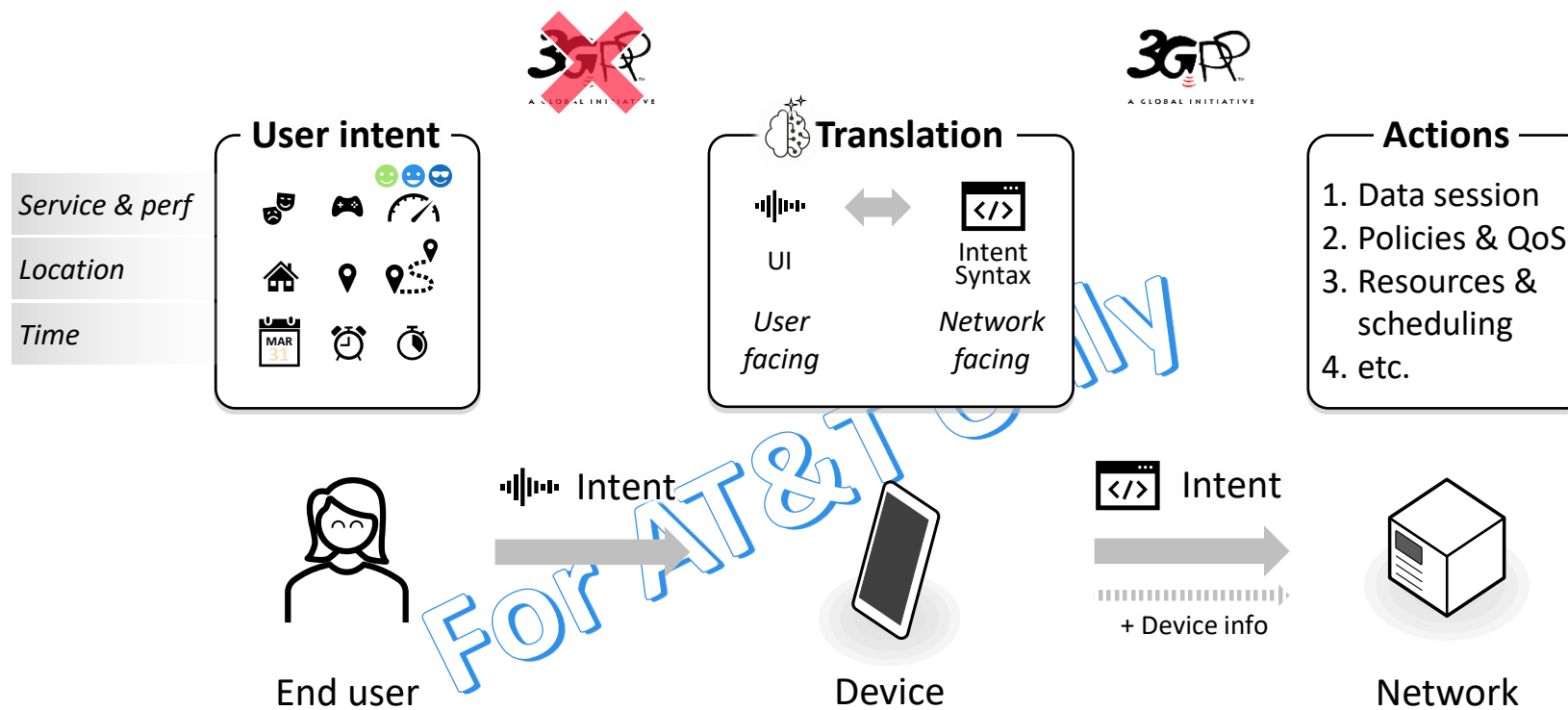
2. Notifying the user of operator feedback / offer



3. Setting up the meeting

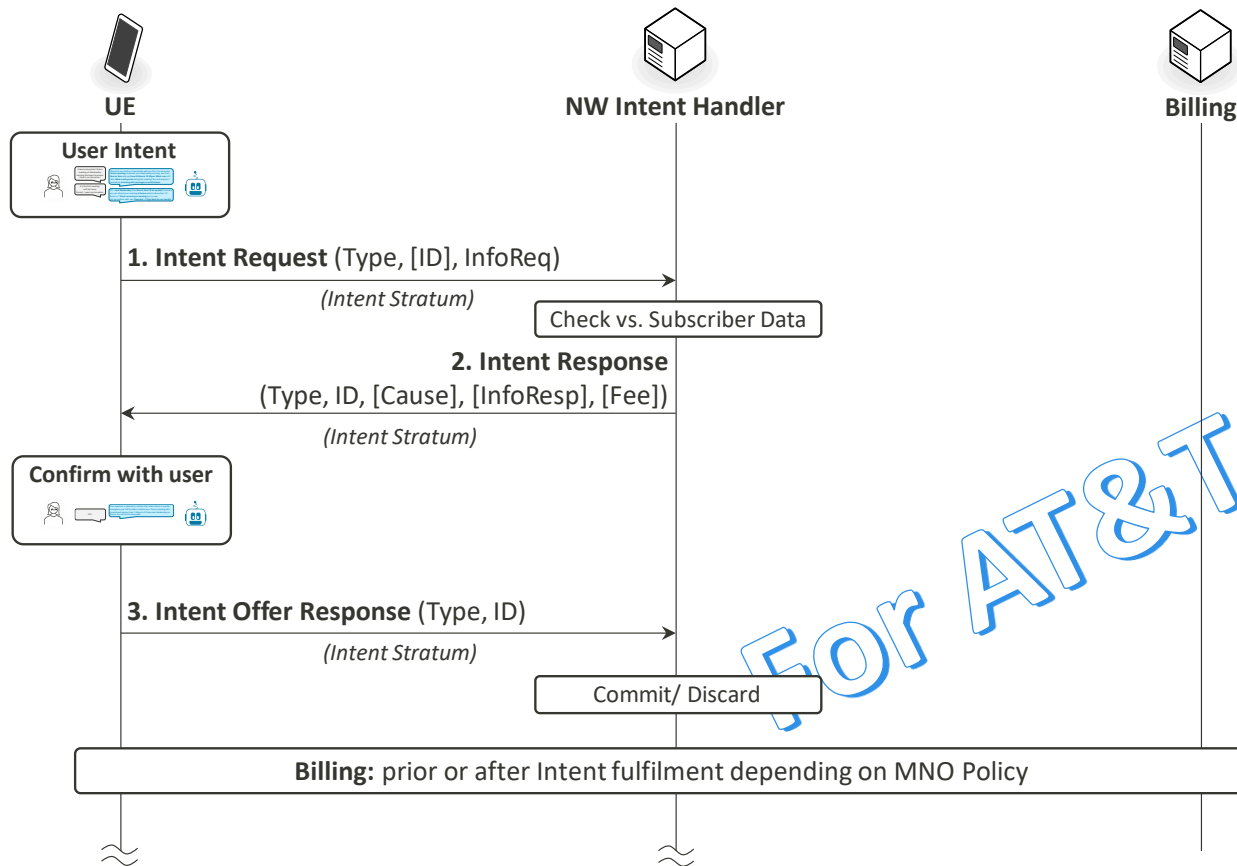
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High-Level Concept



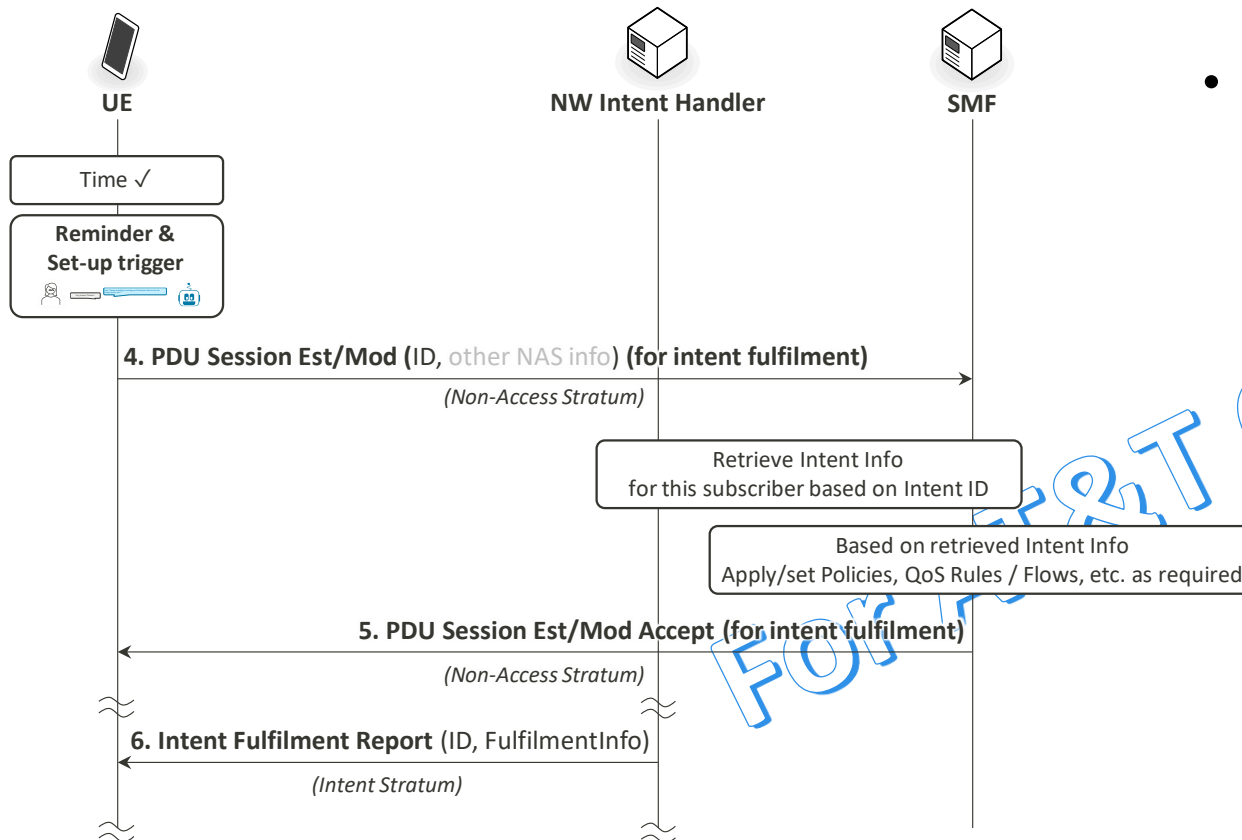
Intent Translation: Natural Language ↔ 3GPP standardized syntax

Intent Delivery and Negotiation



- Intent Request (UE to network)
 - request an intent or the modification of an intent
- Intent Response (network to UE)
 - accept/reject a requested intent (or modification thereof)
 - offer an alternative intent
- Intent Offer Response (UE to network)
 - accept/deny an alternative intent offered by the network

Intent Fulfilment and Notification

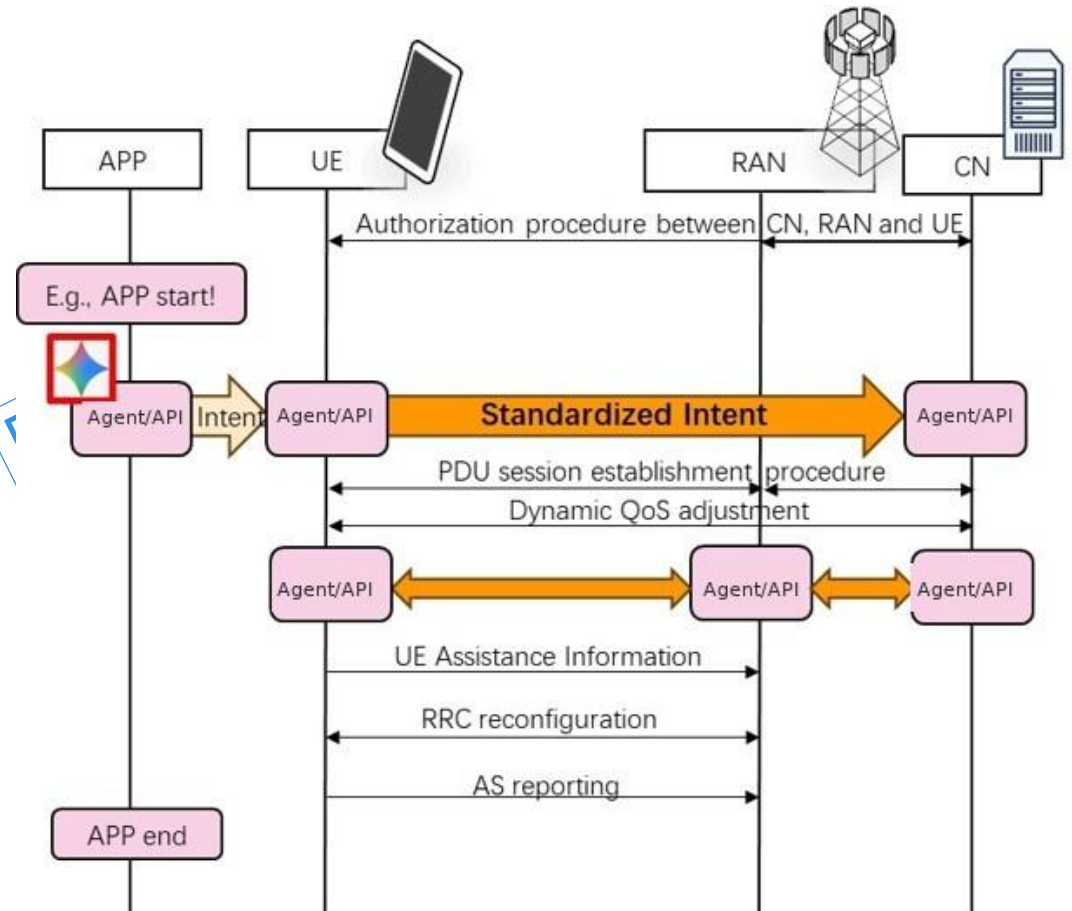


- Intent <> NAS interaction
 - Fulfilment of the negotiated intent via traditional NAS procedures (+intent relation)
 - Fulfilment report to the user

“Intent Stratum”

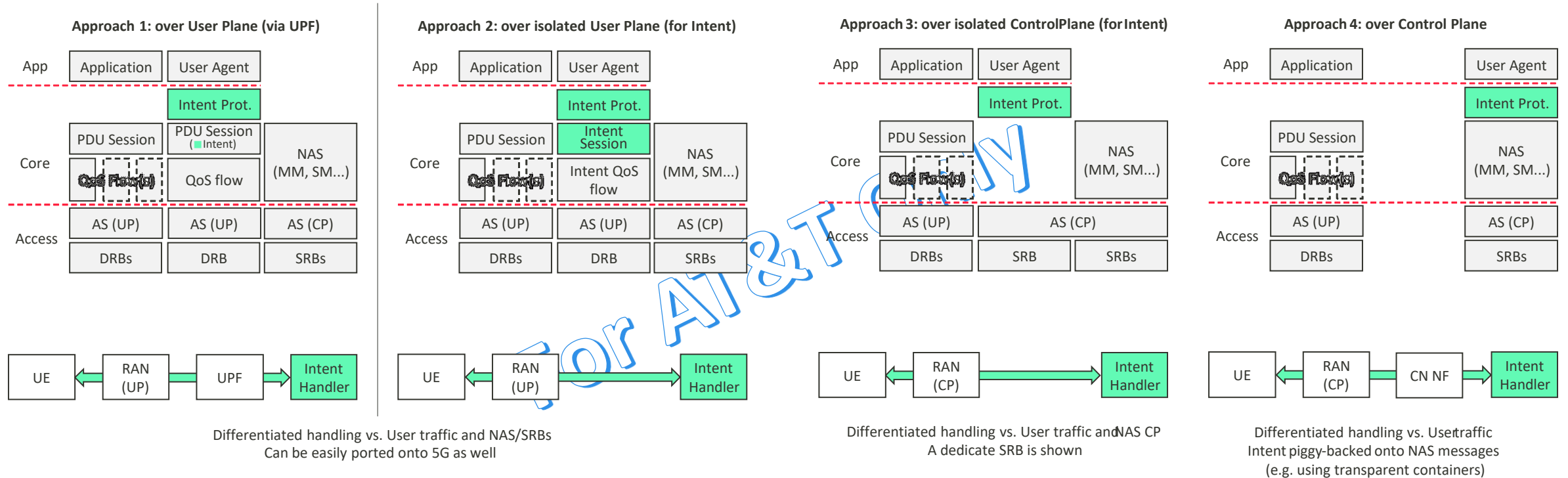
Primarily an editorial convenience (similar to the use of “Stratum” today)

- **Intent Stratum:** Set of protocol(s) between the UE and the NW used for intent transactions over the two sides
 - Operates between the UE and the Intent Handler in the network
 - Could operate between the Intent Handler and other entities (e.g. Network Functions)
 - Note that the “intent stratum” can also be reflected in a system by a set of procedure, APIs, etc., without being explicitly named
- **Intent Handler:** Network entity responsible for handling at least user intent. The Intent Handler could be a dedicated NF, a part of an NF, a network AI agent or subsystem
- **Intent Stratum Definition** can be in a UP or CP manner (next slide)



“Intent Stratum”

Primarily an editorial convenience (similar to the use of “Stratum” today)



3GPP AI-Token/Intent-Based Comm Enablers (1/2)

WG	Area	Feature	Comment
SA2	Intent Interoperability	- Standardized use and expression - Unambiguous interpretation	
	Intent Handler / NW Agent	Parsing and converting intent to network/service requirements	
RAN2	AS Service Awareness	- UE Assistance Information - RRC reconfiguration (intelligent UE config. adjustment) - AS reporting	
	User Plane	- TX latency reduction: CB BSR - reTX latency reduction: SHARQ - Reduce HOL blocking delay: sub-DRB flow - Reduce L2 processing latency: lean protocol stack	
	AI Enhancement	- Traffic prediction - Adaptive RRM/L2 parameters: - TTT - PDCCP discard/reordering timer - Early RLF/BFR/re-establish termination	

3GPP AI-Token/Intent-Based Comm Enablers (2/2)

Area	Feature	Comment
UL Boosting	• DL-UL decoupling • Low-PAPR TC DFT-s OFDM • TX power boosting • UL MIMO augmentation • UE co-MIMO	
Spectral Efficiency	• Overhead minimization • Constellation shaping	
Simplified Mobility	• Fully mobility-free profile for AS and NAS optimizations	
Token/Semantic Layer	• Model split, tokenization, life cycle management	
Intent Stratum	• Intent delivery, negotiation, fulfilment, and AS reporting	

